

PROJECT 1

TASK: DATA CLEANING AND PREPARATION

The messy datasets contain sales records with 14 columns and 1,200 rows. Out of the 1200 rows, there are 309 blanks in the CouponCode column. The objective is to perform data cleaning and preparation for further analysis. The task was accomplished using Excel and Power Query.

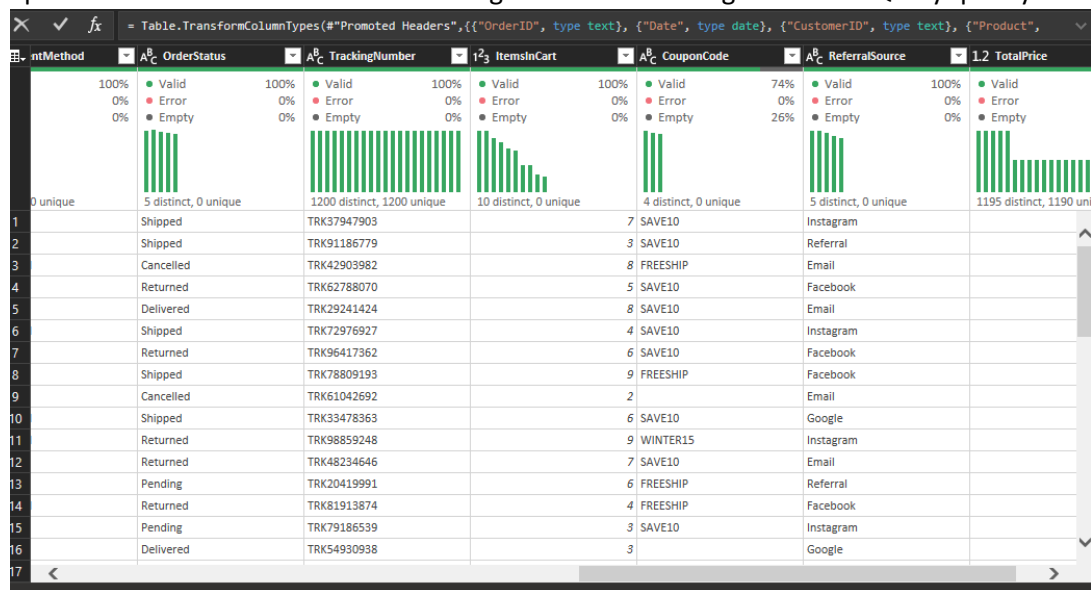
Step 1 – Download and Open with Power Query

- Download the Dataset for the DecodeLab Google Drive (data source)
- Open the downloaded dataset using MS Excel (raw dataset)
- Open another worksheet, and use Get New Data to open the dataset with Power Query in Excel.

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	Shipping	Payment	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD20000	#####	C72649	Monitor	5	570.62	928	Main:Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD20000	#####	C75739	Phone	2	151.35	823	Main:Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD20000	#####	C81728	Tablet	5	550.68	512	Main:Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD20000	#####	C33540	Chair	1	273.19	275	Main:Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD20000	#####	C31840	Printer	4	626.01	668	Main:Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD20000	#####	C37249	Phone	2	245.86	934	Main:Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD20000	#####	C83492	Laptop	1	664.42	986	Main:Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD20000	#####	C41460	Monitor	5	149.55	706	Main:Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD20000	#####	C26817	Phone	2	134.28	904	Main:Gift Card	Cancelled	TRK61042692	2		Email	268.56
ORD20000	#####	C31946	Desk	4	509.38	102	Main:Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD20000	#####	C43443	Tablet	5	625.97	333	Main:Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD20001	#####	C93861	Monitor	3	49.14	831	Main:Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD20001	#####	C38785	Monitor	2	180.5	179	Main:Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD20001	#####	C88348	Laptop	3	201.49	903	Main:Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD20001	#####	C98474	Tablet	2	393.33	980	Main:Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD20001	#####	C39416	Printer	1	473.96	942	Main:Cash	Delivered	TRK54930938	3		Google	473.96
ORD20001	#####	C28976	Printer	2	533.81	914	Main:Credit Card	Pending	TRK131942	3	FREESHIP	Email	1067.62
ORD20001	#####	C60672	Tablet	1	423.4	382	Main:Debit Card	Shipped	TRK119721	3		Referral	423.4
ORD20001	#####	C84329	Desk	1	431.44	891	Main:Gift Card	Pending	TRK6519101	3	SAVE10	Facebook	431.44
ORD20001	#####	C14552	Monitor	5	224	273	Main:Cash	Returned	TRK591015	9	SAVE10	Google	1120
ORD20002	#####	C54829	Chair	5	62.51	587	Main:Cash	Pending	TRK167188	9	WINTER15	Instagram	312.55
ORD20002	#####	C99023	Monitor	4	342.95	569	Main:Cash	Delivered	TRK526465	6		Google	1371.8
ORD20002	#####	C34416	Monitor	2	49.62	633	Main:Online	Pending	TRK75937	2	SAVE10	Google	99.24
ORD20002	#####	C42372	Phone	4	273.25	755	Main:Debit Card	Cancelled	TRK36168	5	FREESHIP	Referral	1093
ORD20002	#####	C17470	Laptop	2	230.95	956	Main:Online	Delivered	TRK28775	3	WINTER15	Referral	461.9
ORD20002	#####	C42577	Monitor	1	399.98	709	Main:Debit Card	Delivered	TRK69423	3	WINTER15	Email	399.98
ORD20002	#####	C21944	Laptop	5	59.55	296	Main:Online	Delivered	TRK80276	6		Referral	297.75
ORD20002	#####	C97593	Printer	1	472.74	881	Main:Online	Shipped	TRK89117	6	WINTER15	Google	472.74
ORD20002	#####	C59823	Phone	4	160.07	568	Main:Online	Pending	TRK65895	5		Email	640.28

Step 2 – Identifying Missing Data

- I profile each column to check for missing or blank data using the Power Query quality bar.



- The **CouponCode** column has 26% empty rows, which is 309 nulls.

- c. To fix this missing data, I replace blanks with “NO COUPON” which the following steps
 - i. Right-click the **CouponCode** column header
 - ii. Click Replace Values
 - iii. In Value to Find — leave it blank (or type null)
 - iv. In Replace With — type **No Coupon**
 - v. Click OK

Step 3 – Removing Duplicates

- a. Select all columns: click the first column header → hold Shift → click the last column header
- b. Go to Home tab > Remove Rows > Remove Duplicates
- c. The row count stays at 1,200 — confirming no duplicates existed

Step 4 – Fixing the Columns Data type

- | | |
|----------------|--------------|
| a. OrderID | Text (ABC) |
| b. Date | Date format |
| c. CustomerID | Text (ABC) |
| d. Product | Text |
| e. Quantity | Whole Number |
| f. Unit Price | \$ Currency |
| g. Total Price | \$ Currency |

Step 5 – Clean and Trim whitespaces from Text Columns

- a. Hold Ctrl and click to select all these columns:
- b. OrderID, CustomerID, Product, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, CouponCode, ReferralSource
- c. Transform tab > Format > Trim
- d. Transform tab > Format > Clean

Step 6 – Apply Standardise Text Case

Apply uppercase to categorical columns:

- a. Hold Ctrl and select: Product, PaymentMethod, and OrderStatus.
- b. Transform tab > Format > Capitalise Each Word

Step 7 - Fix the Date Column

- a. Fix the Date column to comply with ISO 8601
- b. I tried this using the Localise date custom, which does not comply with the standard I need.
- c. After loading the table in Excel, I use date custom
- d. Once loaded into Excel, select the Date column
- e. Press Ctrl + 1 → go to Custom
- f. In the Format box, type: yyyy-mm-dd
- g. Click OK

Step 8 – Load Cleaned Data Into Ms Excel

- a. Go to Home tab > Close & Load.
- b. Click OK
- c. A new worksheet opens with the cleaned dataset.
- d. Rename it to Cleaned Dataset

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	MONITOR	5	570.62	928 Main St	DEBIT CARD	SHIPPED	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75799	PHONE	2	151.35	823 Main St	ONLINE	SHIPPED	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	TABLET	5	550.68	512 Main St	CREDIT CARD	CANCELLED	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	CHAIR	1	273.19	275 Main St	DEBIT CARD	RETURNED	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	PRINTER	4	626.01	668 Main St	ONLINE	DELIVERED	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	PHONE	2	245.86	934 Main St	CREDIT CARD	SHIPPED	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	LAPTOP	1	664.42	986 Main St	GIFT CARD	RETURNED	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	MONITOR	5	149.55	706 Main St	CASH	SHIPPED	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	PHONE	2	134.28	904 Main St	GIFT CARD	CANCELLED	TRK61042692	2	NO COUPON	Email	268.56
ORD200009	2023-11-21	C31946	DESK	4	509.38	102 Main St	CREDIT CARD	SHIPPED	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	TABLET	5	625.97	333 Main St	CREDIT CARD	RETURNED	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	MONITOR	3	49.14	831 Main St	ONLINE	RETURNED	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	MONITOR	2	180.5	179 Main St	DEBIT CARD	PENDING	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C88348	LAPTOP	3	201.49	903 Main St	CREDIT CARD	RETURNED	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	TABLET	2	393.33	980 Main St	DEBIT CARD	PENDING	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	PRINTER	1	473.96	942 Main St	CASH	DELIVERED	TRK54930938	3	NO COUPON	Google	473.96
ORD200016	2023-01-24	C28976	PRINTER	2	533.81	914 Main St	CREDIT CARD	PENDING	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	TABLET	1	423.4	382 Main St	DEBIT CARD	SHIPPED	TRK11972116	3	NO COUPON	Referral	423.4
ORD200018	2025-03-26	C84329	DESK	1	431.44	891 Main St	GIFT CARD	PENDING	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	MONITOR	5	224	273 Main St	CASH	RETURNED	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	CHAIR	5	62.51	587 Main St	CASH	PENDING	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	MONITOR	4	342.95	569 Main St	CASH	DELIVERED	TRK52646983	6	NO COUPON	Google	1371.8
ORD200022	2023-01-29	C34416	MONITOR	2	49.62	633 Main St	ONLINE	PENDING	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	PHONE	4	273.25	755 Main St	DEBIT CARD	CANCELLED	TRK36168083	5	FREESHIP	Referral	1093
ORD200024	2024-10-30	C17470	LAPTOP	2	230.95	956 Main St	ONLINE	DELIVERED	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	MONITOR	1	399.98	709 Main St	DEBIT CARD	DELIVERED	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	LAPTOP	5	59.55	296 Main St	ONLINE	DELIVERED	TRK80276812	6	NO COUPON	Referral	297.75
ORD200027	2024-02-22	C97593	PRINTER	1	472.74	881 Main St	ONLINE	SHIPPED	TRK89117573	6	WINTER15	Google	472.74
ORD200028	2023-11-21	C59823	PHONE	4	160.07	568 Main St	ONLINE	PENDING	TRK65895539	5	NO COUPON	Email	640.28
ORD200029	2024-09-20	C33260	DESK	3	423.53	930 Main St	CREDIT CARD	DELIVERED	TRK59744090	6	SAVE10	Referral	1270.59
ORD200030	2025-03-22	C44267	MONITOR	2	426.7	895 Main St	ONLINE	PENDING	TRK54751444	5	SAVE10	Referral	853.4
ORD200031	2023-06-30	C26445	CHAIR	3	500.08	996 Main St	DEBIT CARD	RETURNED	TRK31229981	6	NO COUPON	Facebook	1500.24
ORD200032	2023-05-01	C12388	TARI FT	5	536.72	830 Main St	CASH	DELIVERED	TRK99261395	10	NO COUPON	Email	2683.6